

highly-skilled profession that is being transformed by AI, radiology also represents a large class of professionals whose main job is a high-stakes classification task. Since we can simulate radiologists’ normal workflow, this is an ideal setting for conducting such an experiment.

While the potential benefits of deploying AI assistance are large in our setting, biases in humans’ use of AI assistance eliminate these potential gains. Even though the AI tool in our experiment performs better than two thirds of radiologists, optimally combining their predictions could substantially improve performance. Yet, we find that giving radiologists access to AI predictions does not, on average, lead to higher performance. This average treatment effect, however, masks systematic heterogeneity: providing AI does improve radiologists’ predictions and decisions for cases where the AI is certain, but not when uncertain. This latter result – that prediction quality can be reduced for some range of AI signals – rejects Bayesian updating with correct beliefs. We also describe systematic errors in belief updating – namely radiologists exhibit automation neglect (e.g. radiologists weight the AI prediction relative to their own) and treat the AI prediction and their own signals as independent even though they are not. Moreover, radiologists take significantly more time to make a decision when AI information is provided.

Together, these results have important implications for how to design the collaboration between humans and machines. Increased time costs and sub-optimal use of AI information both work against having radiologists make decisions with AI assistance. In fact, an optimal delegation policy that utilizes heterogeneity in treatment effects given the AI prediction suggests that cases should either be decided by the AI alone or by the radiologist alone. Only a small share of cases are optimally delegated to radiologists with access to AI. In other words, we find that radiologists should work *next to* as opposed to *with* AI. To the extent that expert decision makers generally under-respond to information other than their own (Conlon et al., 2022) and incorporating additional information is cognitively costly, these insights may hold in other settings where experts’ main job is a classification task.

There are several important considerations that are outside the scope of this work. One question motivated by the unrealized potential gains of AI assistance we find is whether the benefits from AI-specific training for radiologists and/or experience with AI are large. Answering these questions requires different experimental designs or longer-run studies. Another open question is whether the heterogeneity in the value of AI assistance is correlated with a human’s baseline skill or other characteristics, and whether such correlation can be predicted to target assistance. A consideration in the organization of human-AI collaboration that we abstract away from is that the form of collaboration may influence experts’ incentives, to which humans may respond strategically. Finally, the use of AI in practice will be

mediated by organizational incentives and the regulatory environment. Organizations may set guidelines on how to use AI or provide feedback, and regulations may influence liability implications. These issues are interesting avenues for future work.

AI continues to evolve rapidly. This development is increasingly driven by large corporate labs ([Heston and Zwetsloot, 2020](#)), which have resources that are unaffordable to academic institutions. For these and other reasons, economists are unlikely to have a major role in the technical development of AI tools. Our comparative advantage lies in studying how humans interact with these tools and thereby help shape the institutions that guide their use to ensure that this development is beneficial to society. Empirical analysis is a particularly useful tool in this endeavor, especially if the algorithms themselves are a black-box and cannot be understood from first principles.

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